the others. There is always some prophet dressed in white, uttering the words, ‘Follow me, I am the true way.’

I don’t criticize those who prefer to believe in this: we are all free to believe in whatever we want. Maybe, after all, there is a grain of truth in the joke reported by St Augustine: What was God doing before creating the world? He was preparing Hell for those who seek to scrutinize deep mysteries.[2] But these deep mysteries are precisely the ‘depths’ in which Democritus, in the quote that opens this chapter, invites us to seek the truth.

For my part, I prefer to look our ignorance in the face, accept it and seek to look just a bit further: to try to understand that which we are able to understand. Not just because accepting this ignorance is the way to avoid being entangled in superstitions and prejudices – but because to accept our ignorance in the first place seems to me to be the truest, the most beautiful and, above all, the most honest way.

To seek to look further, to go further, seems to me to be one of the splendid things which gives sense to life. Like loving, or looking at the sky. The curiosity to learn, to discover, to look over the next hill, the desire to taste the apple: these are the things which make us human. As Dante’s Ulysses reminds his companions, we are not made ‘to live like brutes, but to seek virtue and knowledge’.

The world is more extraordinary and profound than any of the fables told by our forefathers. I want to go and see it. To accept uncertainty doesn’t detract from our sense of mystery. On the contrary: we are immersed in the mystery and the beauty of the world. The world revealed by quantum gravity is a new and strange one – still full of mystery, but coherent with its simple and clear beauty.

It is a world which does not exist in space and does not develop in time. A world made up solely of interacting quantum fields the swarming of which generates – through a dense network of reciprocal interactions – space, time, particles, waves and light (figure 13.1)

> It continues, it continues, teeming life, and death
> Tender and hostile, clear and unknowable.

And the poet goes on:

> So much the eye can see, from this watching tower.[3]

A world without infinity, where the infinitely small does not exist, because there is a minimum scale to this teeming, beneath which there is nothing. Quanta of space mingle with the foam of spacetime, and the structure of things is born from reciprocal information which weaves the correlations between the regions of the world. A world which we know how to describe with a set of equations. Perhaps, to be corrected.